

Mark Scheme

Q1.

Part	Working or answer an examiner might expect to see	Mark	Notes
	$\lim_{\delta x \rightarrow 0} \sum_{x=4}^9 \sqrt{x} \delta x = \int_4^9 \sqrt{x} \, dx$	B1	This mark is given for writing the expression for a sum as an integral
	$\left[\frac{2}{3} x^{\frac{3}{2}} \right]_4^9 = \frac{2}{3} \times 9^{\frac{3}{2}} - \frac{2}{3} \times 4^{\frac{3}{2}}$	M1	This mark is given for a method to evaluate the integral
	$= \frac{38}{3}$	A1	This mark is given for a correct evaluation of the integral
(Total 3 marks)			

Q2.

Question	Scheme	Marks	AOs
(a)	$x^2 + 8x - 3 = (Ax + B)(x + 2) + C$ or $Ax(x + 2) + B(x + 2) + C$ $\Rightarrow A = \dots, B = \dots, C = \dots$ or $\begin{array}{r} x+6 \\ x+2 \overline{)x^2+8x-3} \\ \underline{x^2+2x} \\ 6x-3 \\ \underline{6x+12} \\ -15 \end{array}$	M1	1.1b
	Two of $A = 1, B = 6, C = -15$	A1	1.1b
	All three of $A = 1, B = 6, C = -15$	A1	1.1b
	(3)		
6(b)	$\int \frac{x^2 + 8x - 3}{x + 2} dx = \int x + 6 - \frac{15}{x + 2} dx = \dots - 15 \ln(x + 2)$	M1	1.1b
	$= \frac{1}{2}x^2 + 6x - 15 \ln(x + 2) \quad (+c)$	A1ft	1.1b
	$\int_0^6 \frac{x^2 + 8x - 3}{x + 2} dx = \left[\frac{1}{2}x^2 + 6x - 15 \ln(x + 2) \right]_0^6$ $= (18 + 36 - 15 \ln 8) - (0 + 0 - 15 \ln 2)$ $= 18 + 36 - (15 - 45) \ln 2$ or e.g. $18 + 36 + 15 \ln \left(\frac{2}{8} \right)$	M1	2.1
	$= 54 - 30 \ln 2$	A1	1.1b
	(4)		
(7 marks)			

Notes:

(a)

M1: Multiplies by $(x + 2)$ and attempts to find values for A, B and C e.g. by comparing coefficients or substituting values for x . If the method is unclear, at least 2 terms must be correct on rhs.

Or attempts to divide $x^2 + 8x - 3$ by $x + 2$ and obtains a linear quotient and a constant remainder.

This mark may be implied by 2 correct values for A, B or C

A1: Two of $A = 1, B = 6, C = -15$. But note that just performing the division correctly is insufficient and they must clearly identify their A, B, C to score any accuracy marks.

A1: All three of $A = 1, B = 6, C = -15$

This is implied by stating $\frac{x^2 + 8x - 3}{x + 2} = x + 6 - \frac{15}{x + 2}$ or within the integral in (b)

(b)

M1: Integrates an expression of the form $\frac{C}{x + 2}$ to obtain $k \ln(x + 2)$.

Condone the omission of brackets around the " $x + 2$ "

A1ft: Correct integration ft on their $Ax + B + \frac{C}{x + 2}$, ($A, B, C \neq 0$) The brackets should be present around the " $x + 2$ " unless they are implied by subsequent work.

M1: Substitutes both limits 0 and 6 into an expression that contains an x or x^2 term or both and a $\ln$ term and subtracts either way round **WITH** fully correct log work to combine two log terms (but allow sign errors when removing brackets) leading to an answer of the form $a + b \ln c$ (a, b and c not necessarily integers)

e.g. if they expand to get $-15 \ln 8 - 15 \ln 2$ followed by $-15 \ln 16$ and reach $a + b \ln c$ then allow the M mark

A1: $54 - 30 \ln 2$ (Apply isw once a correct answer is seen)

Q3.

Question	Scheme	Marks	AOs
(a)	$h = 0.5$	B1	1.1a
	$A \approx \frac{0.5}{2} \{0.5774 + 0.8452 + 2(0.7071 + 0.7746 + 0.8165)\}$	M1	1.1b
	= awrt 1.50	A1	1.1b
	For reference: The integration on a calculator gives 1.511549071 The full accuracy for y values gives 1.504726147 The accuracy from the table gives 1.50475		
		(3)	
(b)	$3 \times \text{their (a)}$ If (a) is correct, allow awrt 4.50 or awrt 4.51 even with no working. Only allow 4.5 if (a) is correct and working is shown e.g. 3×1.5 If (a) is incorrect allow $3 \times \text{their (a)}$ given to at least 3sf but do not be too concerned about the accuracy (as they may use rounded or rounded value from (a)) For reference the integration on a calculator gives 4.534647213	B1ft	2.2a
		(1)	
(c)	This mark depends on the B1 having been awarded in part (b) with awrt 4.5 Look for a sensible comment. Some examples: - The answer is accurate to 2 sf or one decimal place - Answer to (b) is accurate as $4.535 \approx 4.50$ - Very accurate as 4.535 to 2 sf is 4.5 $4.51425 < 4.535$ so my answer is underestimate but not too far off - It is an underestimate but quite close - It is a very good estimate - High accuracy (Quite) accurate - It is less than 1% out $4.535 - 4.5 = 0.035$ so not far out But not just "it is an underestimate" or Calculates percentage error correctly using awrt 4.50 or awrt 4.51 or 4.5 (No comment is necessary in these cases although one may be given) Examples: $\left \frac{4.535 - 4.50}{4.535} \right \times 100 = 0.77\% \quad \text{or} \quad \left \frac{4.535 - 4.51}{4.535} \right \times 100 = 0.55\% \quad \text{or}$ $\left \frac{4.535 - 4.51425}{4.535} \right \times 100 = 0.46\% \quad \text{or} \quad \left \frac{4.50}{4.535} \right \times 100 = 99\%$ In these cases don't be too concerned about accuracy e.g. allow 1sf. This mark should be withheld if there are any contradictory statements	B1	3.2b
		(1)	

(5 marks)

Notes:

(a)

B1: States or uses $h = 0.5$. May be implied by $\frac{1}{4} \times \{ \dots \}$ below.

M1: Correct attempt at the trapezium rule.

Look for $\frac{1}{2}h \times \{0.5774 + 0.8452 + 2(0.7071 + 0.7746 + 0.8165)\}$ condoning slips on the terms but must use all y values with no repeats.

There must be a clear attempt at $\frac{1}{2}h \times (\text{first } y + \text{last } y + 2 \times \text{"sum of the rest"})$

Give M0 for $\frac{1}{2} \times \frac{1}{2} \times 0.5774 + 0.8452 + 2(0.7071 + 0.7746 + 0.8165)$ unless the missing brackets are implied.

NB this incorrect method gives 5.85...

May be awarded for separate trapezia e.g.

$$\frac{1}{4}(0.5774 + 0.7071) + \frac{1}{4}(0.7071 + 0.7746) + \frac{1}{4}(0.7746 + 0.8165) + \frac{1}{4}(0.8165 + 0.8452)$$

May be awarded for using the function e.g. $\frac{1}{2}h \times \left\{ \sqrt{\frac{0.5}{1+0.5}} + \sqrt{\frac{2.5}{1+2.5}} + 2 \left(\sqrt{\frac{1}{1+1}} + \sqrt{\frac{1.5}{1+1.5}} + \sqrt{\frac{2}{1+2}} \right) \right\}$

A1: Awrt 1.50 (Apply isw if necessary)

Correct answers with no working – send to review

(b)

Blft: See main scheme. Must be considering $3 \times$ (a) and not e.g. attempting trapezium rule again.

(c)

B1: See scheme

Q4.

Question	Scheme	Marks	AOs
(a)(i)	$y \times \frac{dx}{dt} = 5 \sin 2t \times 6 \cos t$ or $5 \times 2 \sin t \cos t \times 6 \cos t$	M1	1.2
	$(\text{Area} =) \int 5 \sin 2t \times 6 \cos t \, dt = \int 5 \times 2 \sin t \cos t \times 6 \cos t \, dt$ or $\int 5 \sin 2t \times 6 \cos t \, dt = \int 60 \sin t \cos^2 t \, dt$	dM1	1.1b
	$(\text{Area} =) \int_0^{\frac{\pi}{2}} 60 \sin t \cos^2 t \, dt *$	A1*	2.1*
	(3)		
(a)(ii)	$\int 60 \sin t \cos^2 t \, dt = -20 \cos^3 t$	M1 A1	1.1b 1.1b
	$\text{Area} = [-20 \cos^3 t]_0^{\frac{\pi}{2}} = 0 - (-20) = 20 *$	A1*	2.1
	(3)		
(b)	$5 \sin 2t = 4.2 \Rightarrow \sin 2t = \frac{4.2}{5}$	M1	3.4
	$t = 0.4986\dots, 1.072\dots$	A1	1.1b
	Attempts to find the x values at both t values	dM1	3.4
	$t = 0.4986\dots \Rightarrow x = 2.869\dots$ $t = 1.072 \Rightarrow x = 5.269\dots$	A1	1.1b
	Width of path = 2.40 metres	A1	3.2a
	(5)		
(11 marks)			

Notes:

(a)(i)

M1: Attempts to multiply y by $\frac{dx}{dt}$ to obtain $A \sin 2t \cos t$ but may apply $\sin 2t = 2 \sin t \cos t$ here

dM1: Attempts to use $\sin 2t = 2 \sin t \cos t$ within an integral which may be implied by

e.g. $A \int \sin 2t \times \cos t \, dt = \int k \sin t \cos^2 t \, dt$

A1*: Fully correct work leading to the given answer.

This must include $\sin 2t = 2 \sin t \cos t$ or e.g. $5 \sin 2t = 10 \sin t \cos t$ seen explicitly in their proof and a correct intermediate line that includes an integral sign and the “ dt ”

Allow the limits to just “appear” in the final answer e.g. working need not be shown for the limits.

(a)(ii)

M1: Obtains $\int 60 \sin t \cos^2 t \, dt = k \cos^3 t$. This may be attempted via a substitution of $u = \cos t$ to obtain

$\int 60 \sin t \cos^2 t \, dt = ku^3$

A1: Correct integration $-20 \cos^3 t$ or equivalent e.g. $-20u^3$

A1*: Rigorous proof with all aspects correct including the correct limits and the $0 - (-20)$ and

not just: $-20 \cos^3 \frac{\pi}{2} - (-20 \cos^3 0) = 20$

(b)

M1: Uses the given model and attempts to find value(s) of t when $\sin 2t = \frac{4.2}{5}$. Look for $2t = \sin^{-1} \frac{4.2}{5} \Rightarrow t = \dots$

A1: At least one correct value for t , correct to 2 dp. FYI $t = 0.4986\dots, 1.072\dots$ or in degrees $t = 28.57\dots, 61.42\dots$

dM1: Attempts to find TWO distinct values of x when $\sin 2t = \frac{4.2}{5}$. Condone poor trig work and allow this mark if 2 values of x are attempted from 2 values of t .

A1: Both values correct to 2 dp. NB $x = 2.869\dots, 5.269\dots$

Or may take Cartesian approach

$$5 \sin 2t = 4.2 \Rightarrow 10 \sin t \cos t = 4.2 \Rightarrow 10 \frac{x}{6} \sqrt{1 - \frac{x^2}{36}} = 4.2 \Rightarrow x^4 - 36x^2 + 228.6144 = 0 \Rightarrow x = 2.869\dots, 5.269\dots$$

M1: For converting to Cartesian form **A1:** Correct quartic **M1:** Solves quartic **A1:** Correct values

A1: 2.40 metres or 240 cm

Allow awrt 2.40 m or allow 2.4m (not awrt 2.4 m) and allow awrt 240 cm. **Units are required.**